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GOVERNMENT COLLEGE OF ENGINEERING (AUTONOMOUS), BARGUR

B.E. Degree Examinations – Nov. / Dec. 2023

Fourth Semester / Fifth Semester

ELECTRONICS and COMMUNICATION ENGINEERING

20LPC403/18LPC501 – DIGITAL SIGNAL PROCESSING

Regulations 2018 / 2020

Time : 3 Hours

Maximum Marks : 100

Answer ALL Questions

Part A

(10 x 2 = 20 Marks)

1. List out the classification of discrete time systems.
2. Find the DFT for the signal $x(n) = \delta(n - n_0)$.
3. Find $H(z)$ for the IIR filter whose $H(s) = 1/s + 6$ with $T = 0.1$ sec.
4. What is bilinear transformation? Mention its merits and demerits.
5. Draw the transversal structure for $H(z) = 1 + 2z^{-1} - 3z^{-2} - 4z^{-3}$.
6. Define Gibb's Phenomenon.
7. Why rounding is preferred for quantizing the product?
8. Mention the cause of the three quantization error due to finite word length register in digital filter.
9. Define Decimation and interpolation.
10. What is meant by adaptive filter?

Part B

(5 x 16 = 80 Marks)

11. a) Determine the response of LTI system when the input sequence $x(n) = \{-1, 2, 2, 2, -1\}$ by radix 2 DIT FFT. The impulse response of the system is $h(n) = \{1, 1, -1, 1\}$. (16)

(OR)

- b) Find the response of the output signal $y(n)$ if $h(n) = \{1, 1, 1\}$; $x(n) = \{1, 2, 3, 1\}$ by using linear convolution, Circular convolution and circular convolution with zero padding. (16)

12. a) i) Convert the analog filter with system transfer function (10)

$$H(s) = \frac{(s + 0.1)}{(s + 0.1)^2 + 9}$$

In to a digital IIR filter by means of impulse invariant method.

- ii) Write down the properties of Butterworth filters. (6)

(OR)

- b) i) Realize the following IIR system function in cascade and parallel form (6)

$$y(n) + \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) - 2x(n-1) + x(n-2)$$

- ii) Design a digital Butterworth BPF with passband and stopband attenuations - 3dB and -10dB respectively. The passband cutoff frequencies are 600Hz and 1000Hz and the stopband frequencies are 300Hz and 1600Hz. The sampling frequency is 2000Hz using Bi-linear transformation. (10)



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- ii) Design a digital Butterworth BPF with passband and stopband attenuations - 3dB and -10dB respectively. The passband cutoff frequencies are 600Hz and 1000Hz and the stopband frequencies are 300Hz and 1600Hz. The sampling frequency is 5000Hz using Bi-linear transformation. (10)



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13. a) Design an FIR low pass filter with a cutoff frequency of 1KHz and sampling frequency of 4kHz with 11 samples using Fourier series method. Determine the frequency response and verify the design by plotting the magnitude response. (16)

(OR)

- b) i) Differentiate FIR and IIR filters. (6)
 ii) Design a high pass filter using hamming window with a cut off frequency of 1.2 radians/sec and $N=9$. (10)

14. a) i) The input to the system $y(n) = 0.999y(n-1) + x(n)$ is supplied from an ADC. What is the power reduced by the quantization noise at the output of the filter, if the input is quantized to (a). 8 bits (b). 16bits? (8)
 ii) Determine the characteristics of limit cycle oscillations with respect to the system by the equation $y(n) = 0.95y(n-1) + x(n)$. Find the dead band of the filter. Assume $b = 4$ bits. (8)

(OR)

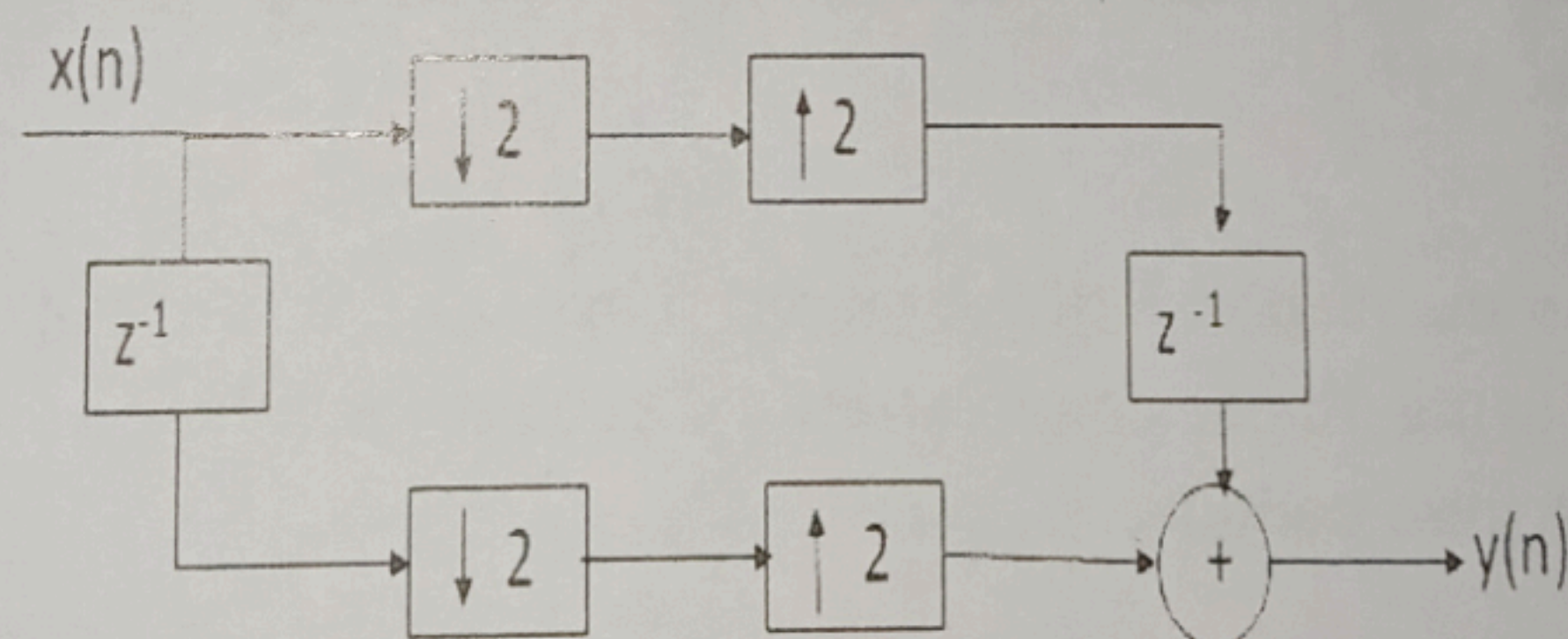
- b) i) Discuss limit cycle oscillations due to roundoff and overflow errors. (8)
 ii) Find the steady state variance of the noise in the output due to quantization of input for the first order filter. (8)

$$y(n) = a \cdot y(n-1) + x(n)$$

15. a) i) Show that the up-sampler and down sampler are time variant system. (8)
 ii) What is a multi-rate system? Write down its applications. (8)

(OR)

- b) i) A multi-rate system is shown in the figure. Find the relation between $x(n)$ and $y(n)$. (8)



$x(n)$ and $y(n)$.

- ii) Explain any two applications of adaptive filtering. (8)

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